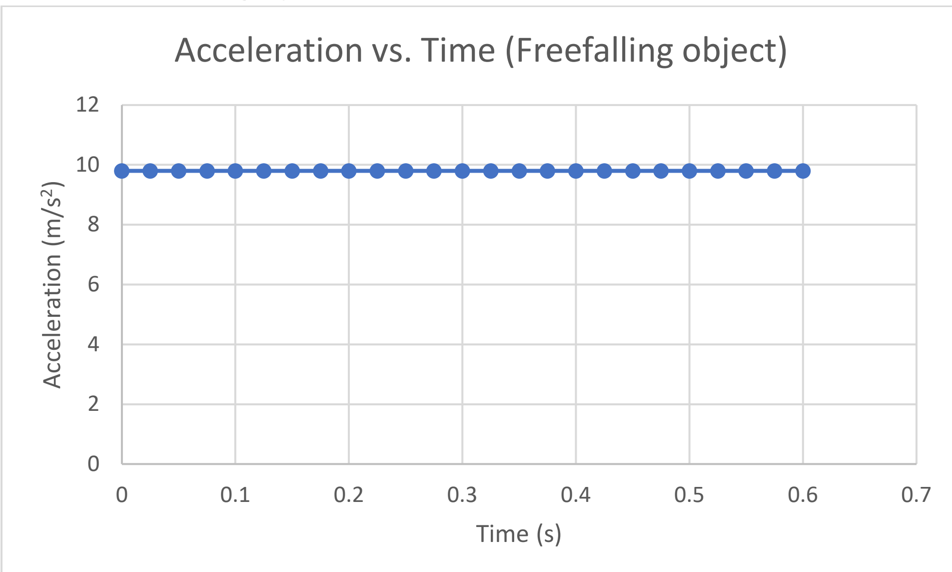
Slope and Area Under the Curve of Motion Graphs

Synopsis: Recall that the slope of the *distance vs. time* graph is the *speed*, and the slope of the *velocity vs. time* graph is the *acceleration*. You will now see that the area under the curve of *acceleration vs. time* is the *velocity change,* and the area under the curve of the velocity vs. time graph is the *distance change*.

Performance Expectations: Calculate velocity change from the area under the curve of acceleration vs. time graph. Calculate distance change from the area under the curve of velocity vs. time graph.

Instructions

1) When an object falls, it accelerates at a constant value of 9.8 m/s^2. The graph below shows the acceleration of this fall (same for *all* objects that fall on Earth) as a function of time (flat line!). **Determine the *area under the curve up to 0.6 s* (the entire time range of the data). Do the calculation in the graph.**



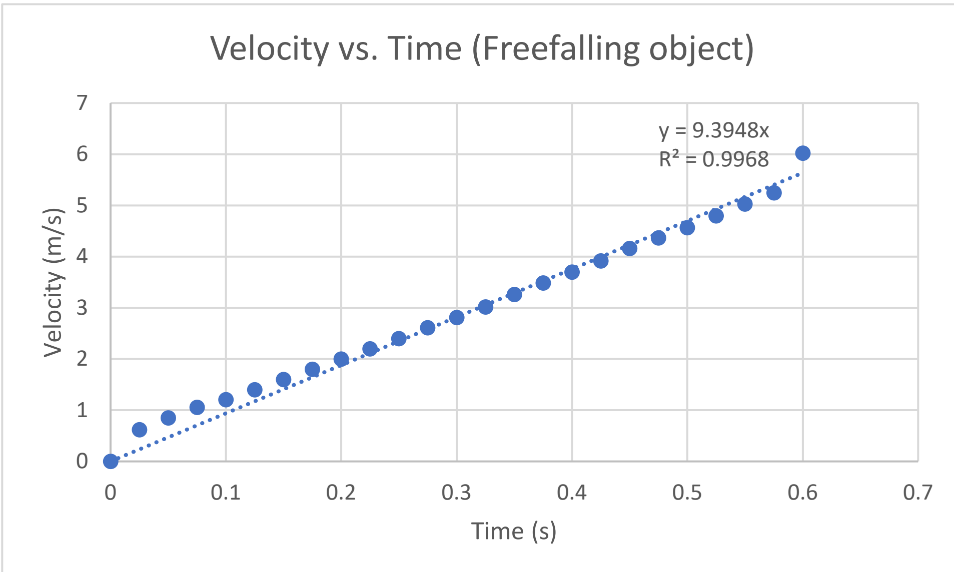
Note, when you take the area under the curve of acceleration vs. time, mathematically, you are doing the following calculation:

Area under the curve = $a \times t = \Delta v$. This comes from none other than $a = \dfrac{\Delta v}{t}$, rearranged for Δv, where $\Delta v = v_f - v_i$, the difference between the final and initial velocity.

Q1) Show that the unit of the area under the curve of acceleration vs. time matches that of the velocity.

2) Below is the velocity vs. time graph of the same measurement. Data points (solid circles) are fitted with the line of best fit, which has a slope of 9.4 m/s^2. You should be wondering if this number ought

to be 9.8 m/s^2. And you are right to be. The difference is attributed to errors stemming from various sources. More on that later.



Q2) Using the line of best fit, estimate the change in the velocity from 0 to 0.6 s.
⊗v = _________ m/s. Does this value match your area under the curve of acceleration vs. time determined above? _______.

3) Using the velocity vs. time graph above, determine the area under the curve up to 0.6 s. Make the triangle using the line of best fit.
Q3) What is the formula for the area of a triangle?

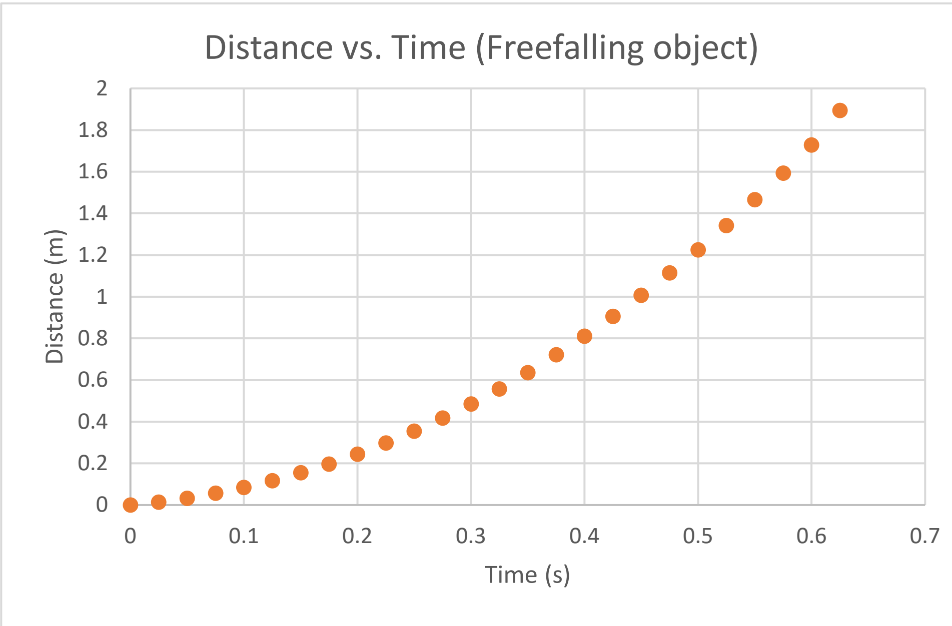
Q4) What is the area under the curve using the formula?

Note, when you take the area under the curve of velocity vs. time, mathematically, you are doing the following calculation:

Area under curve = $\frac{1}{2}\times\Delta v\times t=\Delta d$. This comes from none other than $\Delta v=\frac{\Delta d}{t}$, rearranged for Δd, where $\Delta d = d_f - d_i$, the difference between the final and initial distance. (Don't worry about the factor of ½ - this comes from a constant acceleration. Without acceleration, you don't have a factor of ½). You have not seen the "delta" symbol in front of the distance. Don't worry – this is a minor detail to indicate that the area under the curve is the *change* in distance. d_i, the initial position is almost always zero.

Q5) Show that the unit of the area under the curve of velocity vs. time matches that of the velocity.

4) Below is the distance vs. time graph of the same measurement. Estimate the change in distance from 0 to 0.6 s.



Q6) What is the change in distance from 0 to 0.6 s? $\Delta d =$ ________ m

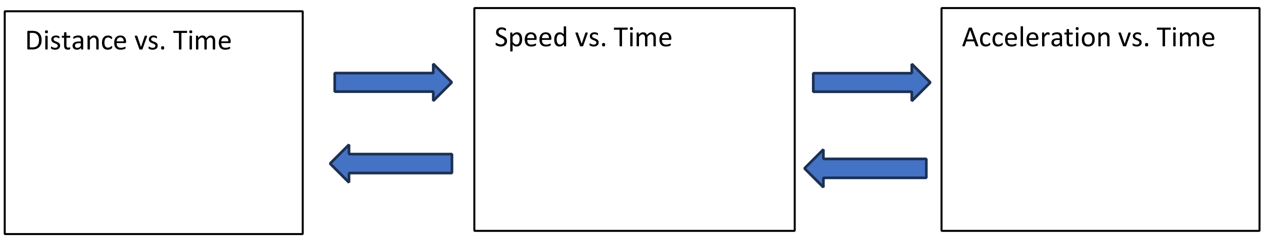
Q7) Does the change in distance in Q6) match that of your estimation from the area under the curve of velocity vs. time graph in Q4)?________.

5) At this point, you should wonder, "So the slope of distance vs. time graph in 4) should produce the velocity vs. time graph in 2)? Absolutely! However, this distance vs. time graph is curved because it's accelerated motion, and you have not come across how to take the slope of a curve. More on this later. We will accept that, like in the constant speed motion, the slope here is the speed.

Summary

1) When you take the slope of the distance vs. time graph, you get s_________d. You know this works because the unit of the slope is _______, and the unit of the sp______ is __________.
2) When you take the area under the curve of velocity vs. time graph, you get d__________e. You know this works because the unit of the area under the curve is _______, and the unit of d_______ is _________.
3) When you take the slope of the velocity vs. time graph, you get ac________. You know this works because the unit of the slope is _______, and the unit of the ac______ is __________.
4) When you take the area under the curve of acceleration vs. time graph, you get s__________d. You know this works because the unit of the area under the curve is _______, and the unit of s_______d is _________.

Label the arrows below with "slope" or "Area under the curve".



<u>In Calculus Notation</u>

The whole ladder is just derivatives going down, and integrals going back up:

$v = dx/dt$ $a = dv/dt = d^2x/dt^2$ (down the ladder)

$\Delta v = \int a\, dt$ $\Delta x = \int v\, dt$ (up the ladder)

Redo your unit checks symbolically using this notation: